

Pharmacodynamic individualization of phage therapy against a KPC-5-producing *Pseudomonas aeruginosa*

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Background: There is a resurgence of interest in bacteriophage (phage) therapy as antimicrobials, resulting from growing antimicrobial resistance to small-molecule antibiotics. Phages are bacterial viruses long studied, but there is a need for high resolution and systematic assessment of clinical dosing strategies for phages to better inform therapy.

Methods: We hypothesized that empirical *in vitro* assessment of clinically relevant phages facilitates pharmacodynamic-driven individualization. Three clinically relevant phage strains (LUZ19, PYO2 and E215) were evaluated as mono- or dual-phage therapy against a clinical *Pseudomonas aeruginosa* in 24 h static time kills and in 7-day hollow fibre infection model.

Results: PYO2 single-bolus administration achieved a bacterial log reduction of 6.82 log₁₀ cfu/mL, with eradication at 4 h. Dual-phage therapy (LUZ19 + PYO2) achieved a bacterial log reduction of 6.81 log₁₀ cfu/mL, with delayed eradication at 12 h.

Conclusions: This highlights the potential of reverse translational pharmacokinetic/pharmacodynamic-driven approaches to guide rational phage selection strategies against individual clinical isolates while identifying potential antagonistic phage-phage interactions.

Introduction

The emergence of extensively drug-resistant *Pseudomonas aeruginosa* (XDR-PA) has been identified by the CDC as a critical public health threat.¹ These challenges have renewed interest in lytic bacteriophages (phages) as alternative therapeutics. Although phages have been used under compassionate-use circumstances, broader clinical translation has been hindered by an incomplete understanding of pharmacokinetic and pharmacodynamic (PK/PD) behaviour. Phages exhibit nonlinear disposition, self-replication at the sites of infection and target-dependent action. As a result, current individualized approaches frequently rely on qualitative assays that may not reflect sustained lytic activity.

Advancing phage therapy beyond individualized use requires quantitative PD frameworks capable of defining exposure-response relationships under clinically relevant conditions. The hollow fibre infection model (HFIM) provides such a platform by recapitulating human phage exposure profiles and enabling

longitudinal assessment of bacterial population dynamics over time. Such PD-driven evaluation is increasingly important for rational cocktail design, as phenotypic diversity and receptor-based complementarity may be insufficient for phage pairing. Establishing PK/PD relationships provides a foundation for identifying phages with true therapeutic potential, and for guiding rational cocktail design and dose characterization for broader therapeutic application.

Methods

A KPC-5-producing *P. aeruginosa* (CDC AR-0231) was used in experimental studies, and a laboratory strain of *P. aeruginosa* (PAO1) was used for phage enumeration using the agar overlay assay. Culturing was performed in cation-adjusted Mueller-Hinton broth or agar (MHA) at 37°C (Becton Dickinson, Sparks, MD, USA). Virulent podovirus LUZ19,² podovirus PYO2³ and myovirus E215³ were freshly propagated and purified prior to treatment using previously established protocols⁴ and stored at 4°C and

protected from light prior to use. These phages were selected based on prior characterization and demonstrated utility in compassionate-use cases.³

Spot assays were performed by applying 4 μ L aliquots of phage stock at concentrations of 10^9 and 10^7 pfu/mL onto bacterial lawns prepared from overnight cultures of each isolate. These inoculum levels were selected to approximate different multiplicities of infections (MOIs) and expected *in vivo* exposures in humans,⁵ accounting for heterogeneity in bacterial burdens observed during infections.^{6–8}

Using an inoculum of 10^7 cfu/mL, time-killing experiments were conducted with LUZ19, E215 and PYO2 using an array of MOIs against CDC AR-0231. The phage MOI array consisted of 10:1, 1:1 and 1:10 of LUZ19, E215 and PYO2 given as monotherapy or as a two-phage cocktail at different MOIs and administration time points. After 24 h of incubation at 37°C, colonies and plaques were enumerated.

For HFIM studies, polyvinylidene fibres (C2025, FiberCell, New Market, MD, USA) with 0.1 μ m pores were inoculated to achieve a starting inoculum of 10^7 cfu/mL. CDC AR-0231 was studied over a 7-day period, with samples collected for measurements of optical density at 620 nm, total bacterial counts and phage concentrations. The system half-life was fixed to 2 h, with a total system volume of distribution of 0.125 L and a clearance rate of 0.0456 L/h, following previously established methods⁹ to obtain a terminal half-life of 2 h. Bolus administration strategies targeted a C_{max} of 7 $\log_{10}$ pfu/mL, achieved with a stock concentration of 11.6 $\log_{10}$ pfu/mL. Continuous infusion strategies targeted a $C_{ss,avg}$ of 2 $\log_{10}$ pfu/mL, achieved with a stock concentration of 6.58 $\log_{10}$ pfu/mL. Bacteria were quantified from samples collected from the extracapillary space of the HFIM cartridge, serially diluted and 50 μ L aliquots were plated onto MHA. Phages were quantified from samples collected from both the central reservoir and extracapillary space of the HFIM cartridge using the agar overlay. Individual phage strains administered in dual-phage regimens were quantified on PAO1-resistant mutant lawns (i.e. PAO1_{LUZ19R}, PAO1_{E215R} and PAO1_{PYO2R}). Triplicate samples were serially diluted in 96-well plates, and 4 μ L aliquots were spotted on MHA pre-lawned with host PAO1 cultured to an OD₆₀₀ ~0.2–0.3. Agar plates were incubated overnight at 37°C before plaque enumeration.

The HFIM enables simulation of human *in vivo* phage exposures by controlling drug administration and PK parameters of the system. Phage samples obtained from the central reservoir reflect the PK, while PD effects are assessed by sampling the extracapillary space, where bacteria are retained. This allows for characterization of the exposure–response relationship by linking the time course of phage exposure to corresponding bacterial counts over time.

Results

At 10^9 pfu/mL, all three phages demonstrated lytic activity against a subset of isolates compared with 10^7 pfu/mL, indicating greater sensitivity at the higher concentration. For CDC AR-0231, lysis was observed with all three phages at 10^9 pfu/mL but at 10^7 pfu/mL, the isolate remained susceptible only to PYO2. Phage titres confirmed the relative MOIs.

In static time kills, PYO2 monotherapy demonstrated significant bactericidal activity, defined as a >99.99% or $\geq 3 \log_{10}$ cfu/mL reduction, with an average decrease of 5.66 $\log_{10}$ cfu/mL (Figure 1a). In contrast, LUZ19 and E215 monotherapy exhibited no bactericidal activity, with bacterial counts resembling those of the untreated growth control. Administration of the 2-phage cocktail (PYO2+LUZ19) produced diminished bactericidal activity relative to PYO2 monotherapy, achieving an average reduction of 3.16 $\log_{10}$ cfu/mL (Figure 1b). Consistent with this observation, the nadir bacterial burden achieved with the PYO2 and LUZ19

combination was significantly higher than that achieved with PYO2 monotherapy, with a mean difference of 2.53 $\log_{10}$ cfu/mL ($P < 0.001$), supporting a reproducible antagonistic effect. However, the two-phage cocktail containing E215 and LUZ19 exhibited no bacterial killing, consistent with the inactivity observed in each monotherapy. Phage titres corresponding to static time kill experiments are provided in Figure S1 (available as [Supplementary data](#) at JAC-AMR Online).

To further investigate phage–phage interactions between dual-phage therapy, different combination regimens, including both simultaneous and sequential administration, were tested against CDC AR-0231 (Figure 1c and d). Within the two-phage cocktail, different MOI ratios of different phages were explored (Figure 1e). PYO2 (MOI 10:1)+LUZ19 (MOI 1:1) and PYO2 (MOI 1:1)+LUZ19 (MOI 10:1) exhibited similar bactericidal activity, resulting in bacterial reductions of 3.55 and 2.97 $\log_{10}$ cfu/mL, respectively. Despite differences in relative abundance of individual phages, bacterial growth profiles remained nearly identical, suggesting that individual phage concentrations have minimal impact on bactericidal activity against CDC AR-0231. Additional two-phage regimens using E215 and LUZ19 also failed to produce bacterial killing across experimental conditions.

Sequential administration of PYO2 and LUZ19 (Figure 1c) involved culturing one phage at 0 h, followed by the addition of a second phage at 2 h at an MOI of 1:1. When LUZ19 was introduced first (0 h), bacterial counts mirrored those of the untreated growth control. However, subsequent addition of PYO2 (2 h) led to a 4.02 $\log_{10}$ cfu/mL bacterial reduction by 4 h. In contrast, when PYO2 was administered first (0 h), bacterial counts decreased by 4.33 $\log_{10}$ cfu/mL at 2 h.

Results from the HFIM are summarized in Figure 2. Monotherapy control arms for each individual phage were evaluated as either bolus dosing or continuous infusion. Bolus dosing was designed to emulate *in vivo* phage concentrations, achieving an MOI of 1:1.⁵ Using linear regression with samples collected at 2, 4, 6, 8 and 12 h, peak phage concentrations in the central reservoir were estimated at 6.93, 6.98 and 7.00 $\log_{10}$ pfu/mL for PYO2, E215 and LUZ19, respectively. Monotherapy with E215 or LUZ19, administered via bolus dosing, exhibited no bactericidal activity, with bacterial counts closely resembling those of the untreated growth control. However, PYO2 bolus monotherapy resulted in a minimum count of 0 $\log_{10}$ cfu/mL by 2 h, with sustained suppression until 28 h. Continuous infusion regimens resulted in reduced bacterial killing for PYO2, while E215 and LUZ19 continued to exhibit no bactericidal activity. PYO2 administered via continuous infusion resulted in delayed bactericidal activity exhibiting a reduction of 1.54 $\log_{10}$ cfu/mL between 48 and 56 h, despite phage concentrations in the central reservoir consistently exceeding the steady-state threshold of 2 $\log$ pfu/mL. Additionally, PYO2 concentrations in the extracapillary space peaked at 10.51 $\log_{10}$ pfu/mL following bolus administration and 9.86 $\log_{10}$ pfu/mL under continuous infusion.

Two-phage cocktail regimens were evaluated as bolus therapy (MOI 1:1) to assess phage–phage interactions. The combination of E215+LUZ19 exhibited no bactericidal activity throughout the course of treatment, mirroring single-phage therapy and indicating a lack of synergy. Comparatively, single bolus dosing of PYO2+LUZ19 achieved undetectable counts at 12 h followed by rapid regrowth, suggesting potential antagonism with the addition of LUZ19.

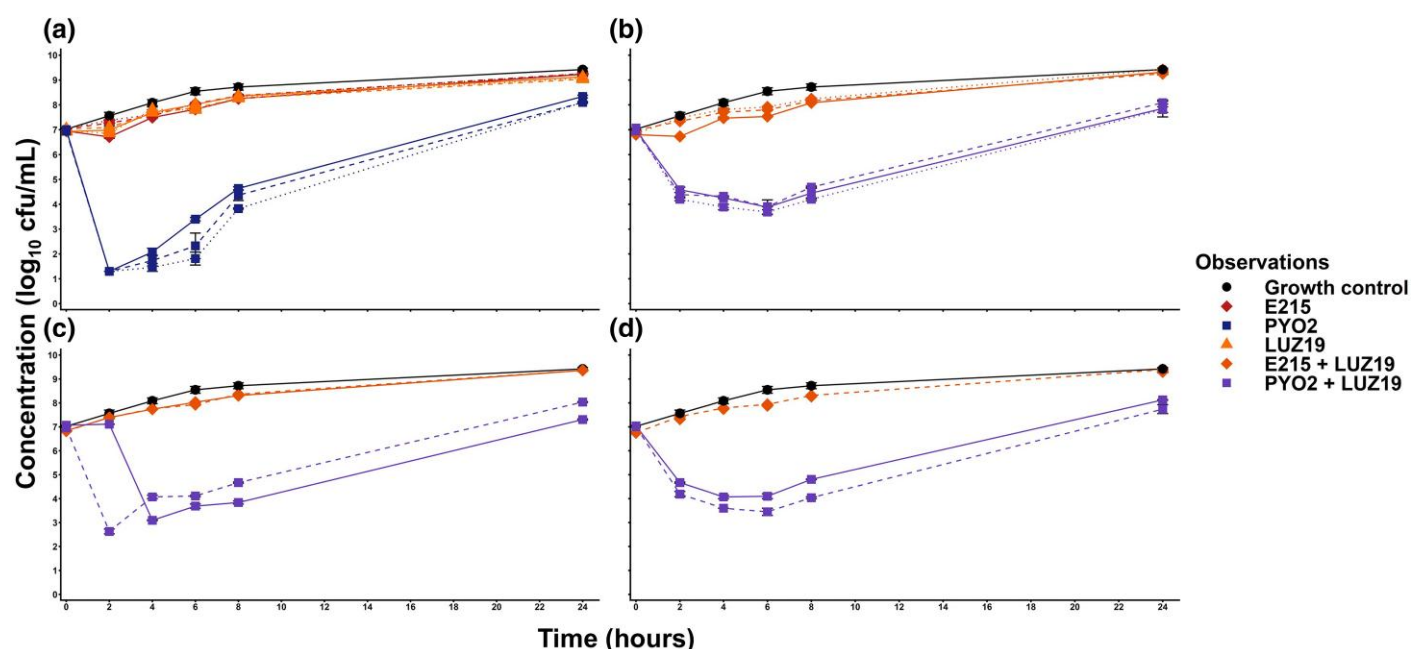


Figure 1. Potential phage–phage interactions against CDC AR-0231 during simultaneous and sequential administration of dual-phage therapy. Bacterial counts are reported as $\log_{10}$ cfu/mL (mean $\pm$ standard error). Colour and symbol denote treatment arm, while line type denotes MOI or dosing sequence. (a) Phage monotherapy evaluated at MOIs of 10:1 (dotted), 1:1 (dashed) and 1:10 (solid). PYO2 monotherapy exhibited rapid bactericidal activity at each MOI, reducing bacterial counts to the limit of detection within 2 h. LUZ19 and E215 monotherapy had no measurable effect compared with the growth control. (b) Simultaneous dual-phage therapy evaluated at MOIs of 10:1 (dotted), 1:1 (dashed) and 1:10 (solid). Regimens with LUZ19 and PYO2 produced significant bactericidal activity at each MOI but regimens with LUZ19 and E215 resulted in minimal bacterial reduction. (c) Sequential dosing at MOI 1:1, with one phage administered at 0 h followed by the second phage at 2 h. Administration of PYO2 followed by LUZ19 (dashed) resulted in rapid bactericidal activity, reaching a nadir of 2.63 $\log_{10}$ cfu/mL by 2 h, whereas administration of LUZ19 followed by PYO2 (solid) initially mirrored the growth control, with bactericidal activity emerging only after PYO2 introduction, resulting in a 4.02 $\log_{10}$ cfu/mL bacterial reduction by 4 h. In contrast, sequential dosing with E215 and LUZ19 produced no measurable bacterial reduction relative to growth control. (d) Simultaneous administration of PYO2 and LUZ19 at reciprocal MOIs (10:1 and 1:1) resulted in comparable bactericidal activity regardless of relative phage exposure, whereas E215 and LUZ19 exhibited no lytic activity under either condition.

Discussion

This study demonstrates that quantitative PD assessment is indispensable for identifying phages with true therapeutic potential against extensively drug-resistant isolates. Although spot assays suggested that CDC AR-0231 was broadly susceptible to three candidate phages, only PYO2 produced meaningful lytic activity when evaluated under static and dynamic conditions. In contrast, LUZ19 and E215 exhibited minimal activity despite positive qualitative screening. This disconnect highlights that qualitative assays alone are not sufficient to guide individualized phage therapy under compassionate-use circumstances.^{10–12}

Excluding phages that fail initial qualitative screens inherently constrains the discovery space for phage–phage interactions. As a result, potential antagonistic pairings are systematically excluded before they can be assessed, creating an implicit expectation of synergy. Under such assumptions, the antagonism observed between LUZ19 and PYO2 in our study would have remained undetected. Qualitative screenings therefore not only fall short in predicting lytic activity but also predetermines which phage–phage interactions can be discovered.

While isolate-to-isolate variability in phage susceptibility is well recognized, our findings highlight that phage–phage interactions may vary across isolates. We previously demonstrated that

LUZ19 exhibited synergistic activity with both PYO2 and E215 in the HFIM against a reference isolate of *P. aeruginosa*,^{9,13} yet against CDC AR-0231 the same phage diminished the activity of PYO2. These divergent outcomes suggest that the behaviour of a phage in combination cannot be predicted from its individual lytic activity or from interactions observed in other isolates.

Across a three-log range of MOIs, bacterial reduction remained largely unchanged, suggesting that lytic activity may follow a threshold-like behaviour rather than dose-proportionate effects.¹⁴ These findings challenge assumptions surrounding the use of high MOIs, where dosing is often guided by empirical convention rather than defined exposure–response relationships. By recapitulating expected *in vivo* phage exposures, the HFIM provides a translational platform to evaluate these population dynamics under clinically relevant conditions.

Together, these observations should be interpreted in the context of several limitations. A primary limitation of this work is the use of a single XDR *P. aeruginosa* isolate, which restricts broader applicability of the findings. Because XDR infections arise from genetically diverse strains, defining the therapeutic potential of our candidate phages will require evaluation across a large panel of clinical isolates. Such breadth is necessary to understand how individual phages and phage cocktails behave across epidemiological diversity to determine whether a given cocktail

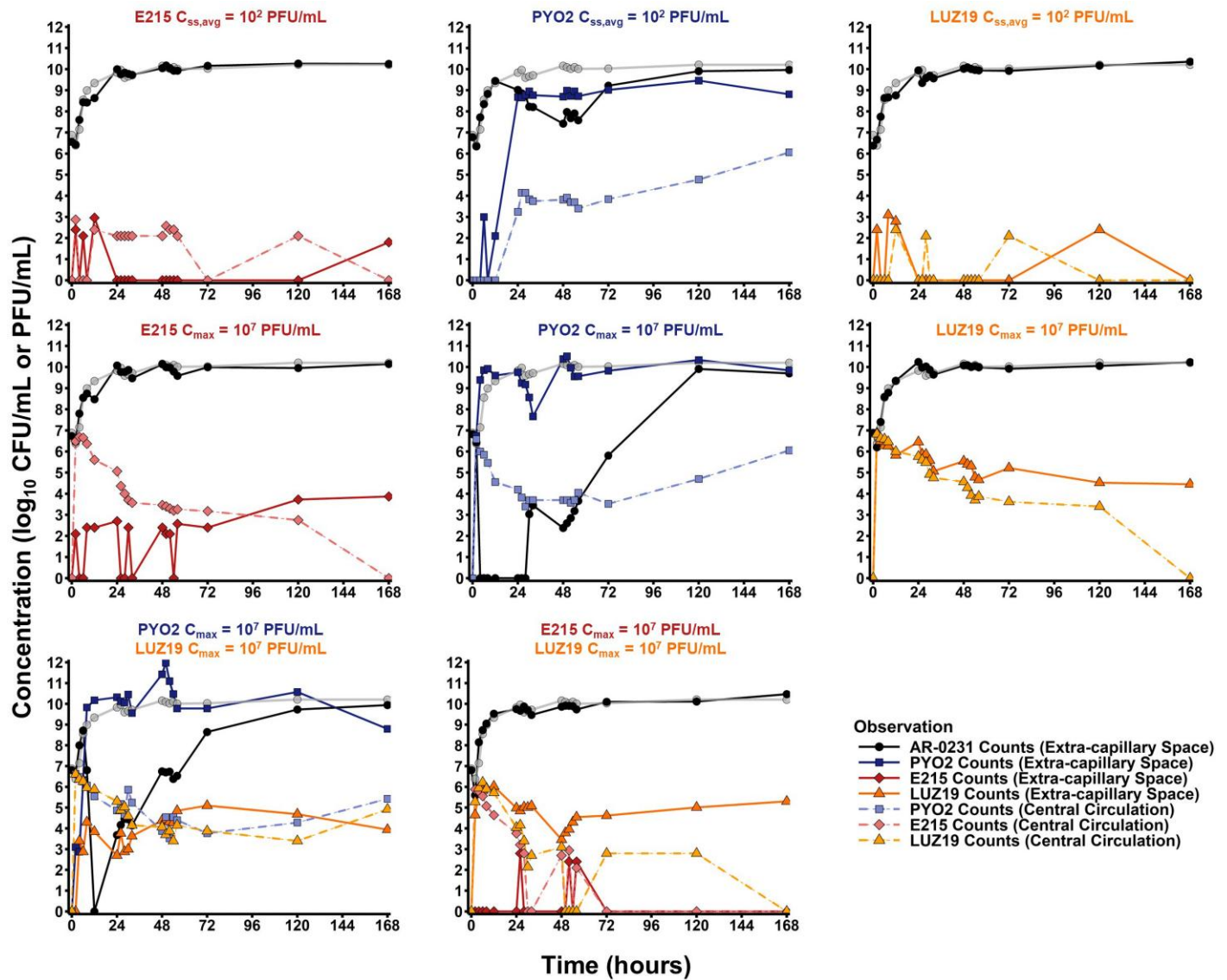


Figure 2. Hollow fibre infection model results. Total observed counts for CDC AR-0231 for each regimen tested are overlaid with the growth control. Total counts in the extracapillary space/central circulation for E215, PYO2 or LUZ19 are reported with relevant experiments.

can achieve meaningful coverage. Designing a broadly active cocktail therefore depends on frameworks that encompass the diversity of circulating XDR isolates and the variation in interaction profiles that emerge across them. Phage resistance emerged during prolonged HFIM exposure, which is expected under sustained selective pressure, but the mechanisms underlying resistance were not explored. Similarly, the mechanism driving the antagonistic interaction between LUZ19 and PYO2 was not examined, despite these phages having been previously combined³ on the basis of complementary phenotypic diversity to minimize potential antagonism.^{15,16} Notably, PYO2 monotherapy consistently achieved a greater bacterial log reduction than dual-phage therapy, indicating a reproducible attenuation of bactericidal activity with the addition of LUZ19. Future work

incorporating multiple XDR isolates and mechanistic studies of resistance and inter-phage interactions will be important for refining phage selection and optimizing cocktail design.

Efforts to optimize phage therapy increasingly emphasize deliberate cocktail design, yet selection strategies based solely on host-range profiles or receptor diversity provide an incomplete basis for predicting inter-phage interactions. Quantitative PD approaches offer a more informative framework by capturing interaction patterns, defining exposure-response relationships, and identifying combinations with therapeutic potential. Expanding these evaluations across diverse XDR *P. aeruginosa* isolates, coupled with dual *in vitro*-*in silico* approaches, highlights the potential of reverse translational PK/PD-driven strategies to guide rational phage dosing and cocktail design.⁹ Prospective studies

exploring phage-antibiotic synergy, particularly the mechanistic basis underlying these interactions, may broaden opportunities for cocktail development. Ultimately, PD-driven strategies offer a path towards standardized and individualized phage therapies for resistant infections.

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Transparency declarations

None to declare.

Supplementary data

Figure S1 is available as Supplementary data at [JAC-AMR](#) Online.

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